Machine Learning

Kapil Dev Pant (6 hrs)

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Contents

1. Definition and Evolution of Machine Learning
2. Learning by Analogy
3. Explanat ion-based learning
4. Supervised Learning Algorithms
    1. Classification and Regression
    2. Linear regression
    3. K-Nearest Neighbor
5. Unsupervised Learning Algorithms
    1. Clustering
    2. K-means Clustering
6. Artificial Neural Network
    1. Biological Inspiration
    2. Basic Components of ANN
    3. Training Neural Networks
        1. Forward Propagation
        2. Loss Function
        3. Backward Propagation
        4. Learning Rate
    4. Single-Layer Perceptron
    5. Multi-Layer Perceptron

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Machine Learning

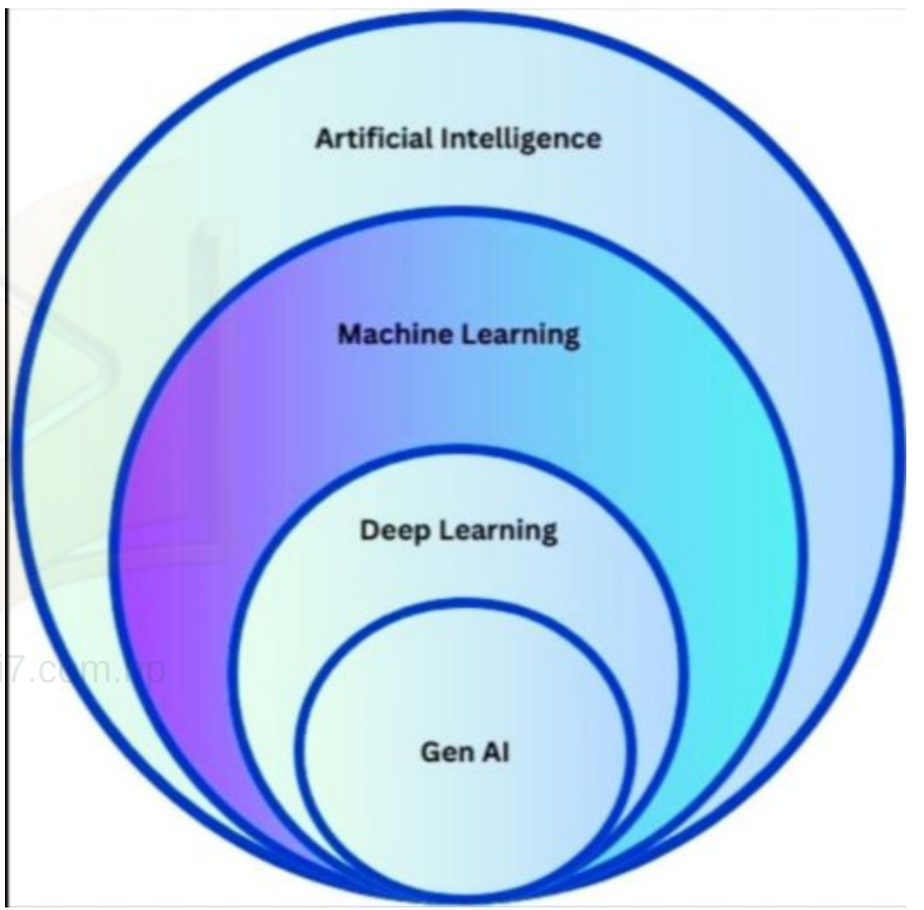
- What is learning and why needed?
- Why we want agent to learn? | we just improve the ploblem that is there in model?
  - Designer can't anticipate all possible situation, that agent might come across. e.g.: maze
  - Designer can't anticipate all changes over time. Eg : stock market
  - Human programmer don’t have idea to solve problem by themselves. Eg: face detection

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Machine Learning

- Field of Artificial Intelligence (AI) that enables systems to learn and improve from experience without being explicitly programmed.
- Focus on developing algorithms that can process data, identify patterns, and make decisions.
- Key characteristics
  - Data Driven Approach
  - Adaptive and Scalable
  - Predictive Modeling



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Evolution of Machine Learning

1. 1950s: Concept of learning machines introduced (Alan Turing’s "Learning Machine").
2. 1960s-70s: Symbolic AI and simple ML algorithms developed (e.g., Perceptron).
3. 1980s-90s: Neural networks and support vector machines (SVMs) emerged.
4. 2000s: Rise of deep learning and big data.
5. 2010s-Present: Advanced techniques like reinforcement learning, GANs, and real-world applications.

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Learning by Analogy

- Learning by drawing parallels between a new situation and previously encountered situations.
- Solving a problem in one domain by applying the solution from a different domain.
- Involves abstracting details from a particular set of problems and resolving structural similarities between previously distinct problems

Applications:

- Case-based reasoning systems
- Natural language processing (e.g., metaphor understanding)

Advantages:

- Intuitive and efficient for human-like learning.

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Explanation-Based Learning (EBL)

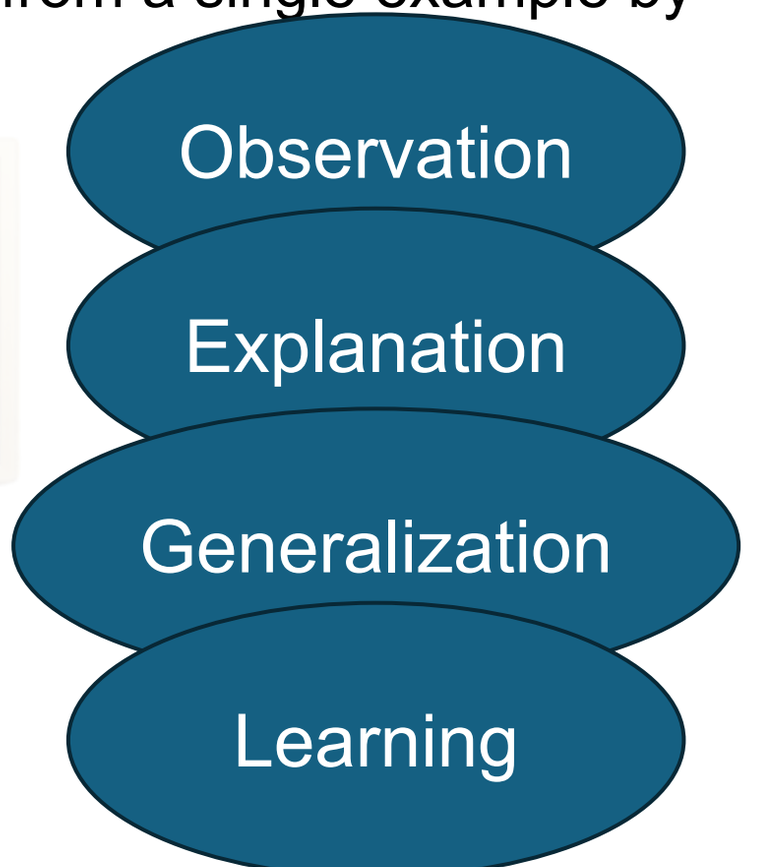
A learning technique where a system generalizes from a single example by understanding the underlying explanation.

Steps in EBL:

1. Analyze a specific example.
2. Extract the generalized rule or explanation.
3. Apply the rule to similar problems.

Applications:

- Knowledge-based systems.
- Robotics and automated planning.



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Feedback Based

- Supervised Learning
- Unsupervised Learning
- Reinforced Learning

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Supervised Learning

Learning from labeled data, where the algorithm maps inputs to known outputs.

Key Characteristics:

- - Requires a labeled dataset.
- - Goal: Minimize the error between predictions and actual outputs.

Applications:

- - Email Spam Classification
- - Stock Price Prediction
- - Image Recognition

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Two approaches of Supervised Learning

1. Classification: A classification problem is when the output variable is a category, such as “Red” or “blue” , “disease” or “no disease”.

    Example: Classifying emails as spam or not spam.

    Algorithms: Logistic Regression, Decision Trees, SVM, Random Forests, Naive Baye.

2. Regression: A regression problem is when the output variable is a real value, such as “dollars” or “weight”.

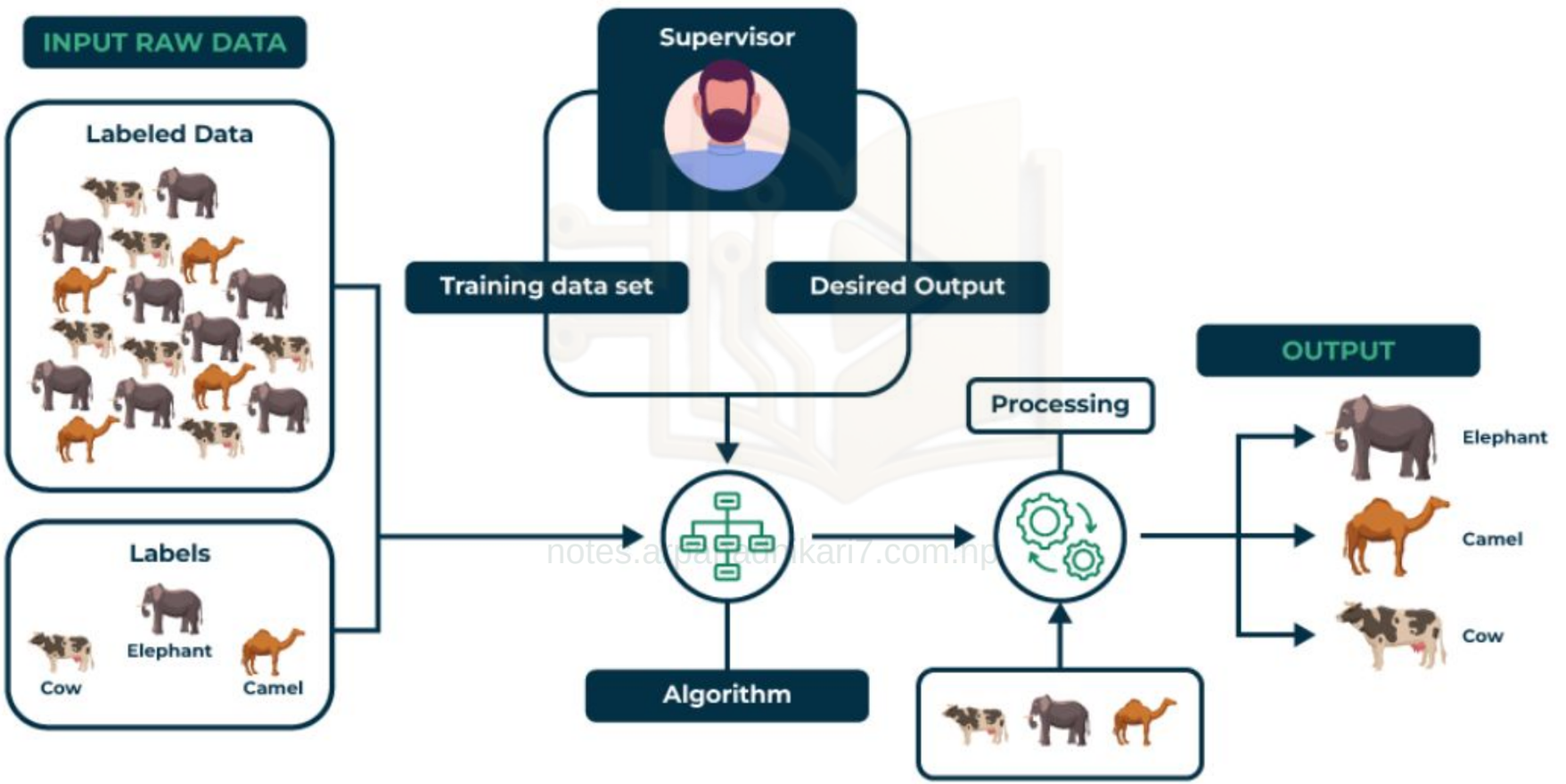
    Example: Predicting house prices.

    Algorithms: Linear Regression, Ridge Regression, Support Vector Machine Regression, Decision Tree Regression, Random Forest Regression.

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Supervised Learning



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Types of Supervised Learning Algorithms

- Classification
- Classification is a technique with which we can **affiliate** data under certain labels. We can train algorithms and models to classify data under different classes.
- For example, if you have data of fruits consisting of apples and oranges, your model will learn from the data about the specifications of the fruits.
- The next time you enter new data, the model will compare your data and then classify it under either class.

    Binomial Classification

    Multi-Class Classification
- Regression
- Regression helps us to understand the relationship between various data points and helps us to find hidden patterns among the data.
- We have both dependent and independent variables in this case which can help us to understand the relation with the help of **graphical** representations.

    Linear Regression
    Logistics Regression

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K nearest Neighbor (KNN)

- A non-parametric algorithm that classifies or predicts a data point based on the majority class (or average value) of its k-nearest neighbors.
- Working:
  - Choose the number of neighbors (‘k’).
  - Calculate distances between the data point and all other points.
  - Identify the k-nearest neighbors.
  - For classification: Assign the majority class.
  - For regression: Compute the average of neighbors.
- Distance Matrices
  - Euclidean
  - Manhattan
  - Minkowski

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Application of Supervised Learning

- prediction of stock markets
- Speech Recognition
- Object detection

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Unsupervised Learning

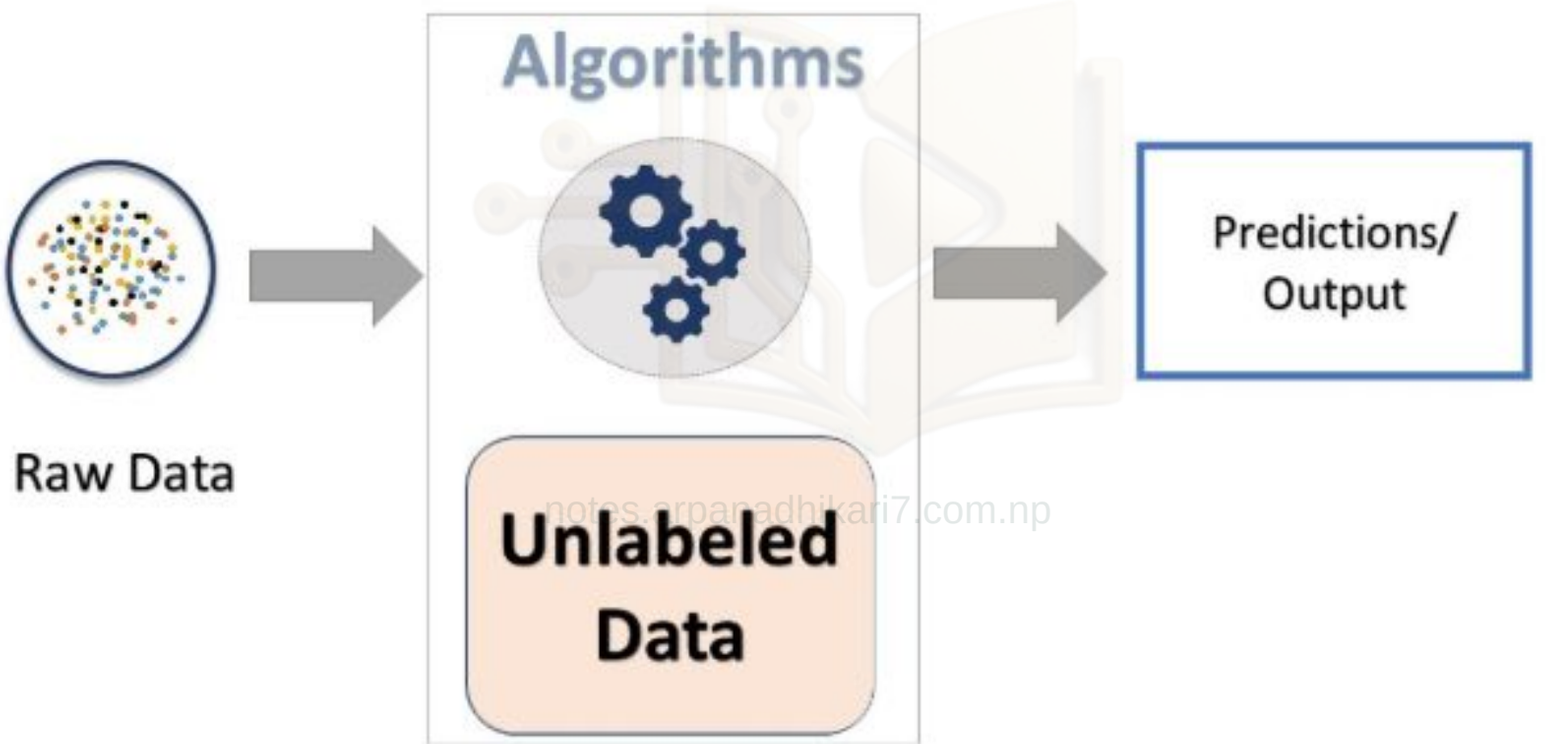
Learning from unlabeled data, identifying patterns and structures.

- Key Characteristics:
- - No labeled outputs.
- - Goal: Discover hidden structures or features in data.
- Applications:
- - Customer Segmentation
- - Fraud Detection
- - Dimensionality Reduction

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Unsupervised Learning



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Types of Unsupervised learning

1. Clustering:
- - Example: Grouping customers by purchasing behavior.
- - Algorithms: K-Means, Hierarchical Clustering.
2. Dimensionality Reduction:
- - Example: Reducing the number of features in a dataset.
- - Algorithms: PCA, t-SNE.

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Clustering

- An unsupervised learning task that groups data points into clusters based on similarity.
- Types of Clustering:
  - Centroid-based: K-means
  - Density-based: DBSCAN
  - Hierarchical: Agglomerative Clustering
- Applications:
  - Customer segmentation
  - Document categorization

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K Means Clustering

- A centroid-based clustering algorithm that divides data into k clusters by minimizing within-cluster variance.
- Algorithm Steps:
  - Choose the number of clusters (‘k’).
  - Initialize ‘k’ centroids randomly.
  - Assign each data point to the nearest centroid.
  - Update centroids by computing the mean of assigned points.
  - Repeat steps 3 and 4 until convergence.
- Use Cases:
  - Market segmentation
  - Image compression

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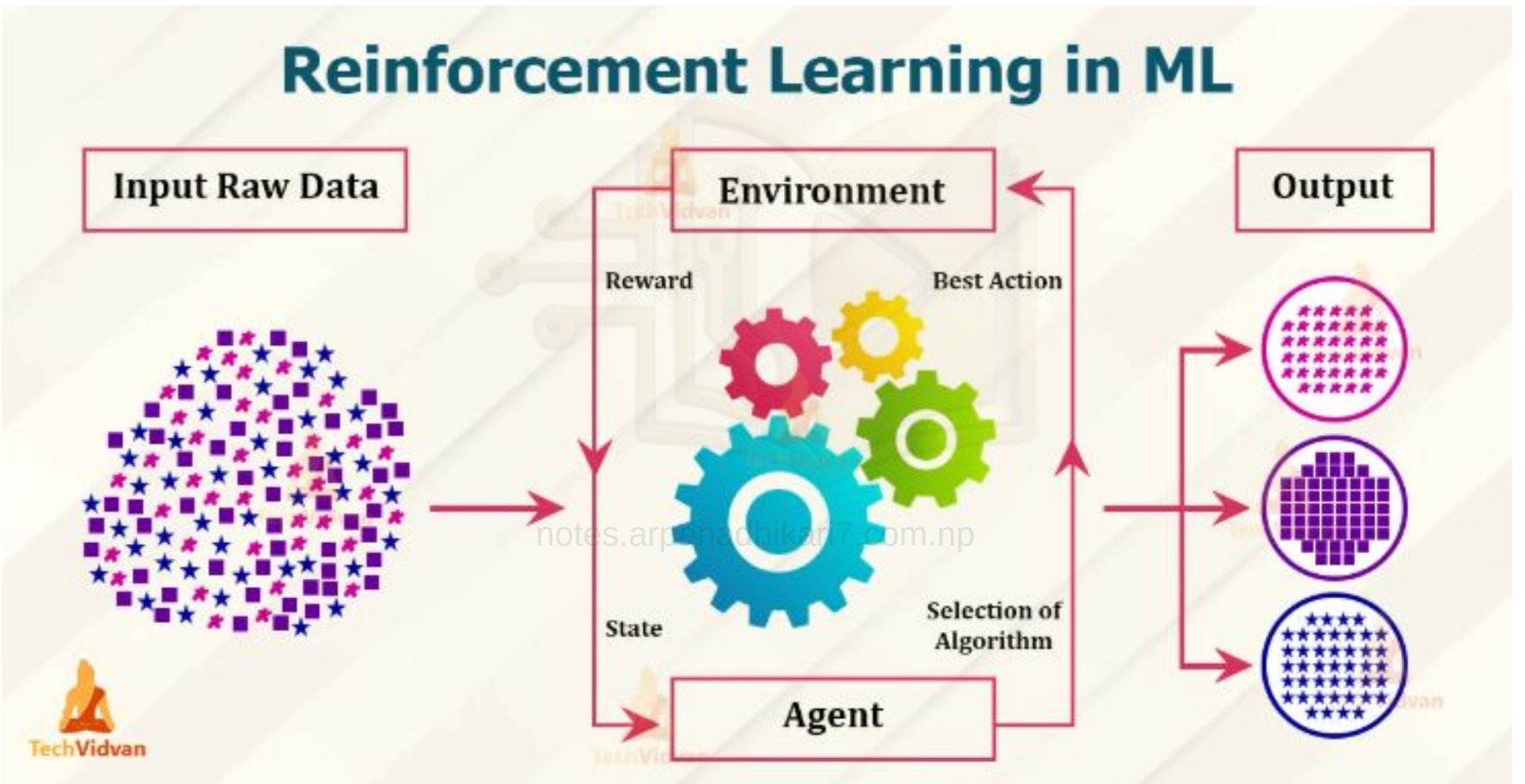
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Reinforcement Learning

- Definition: Learning by interacting with an environment to maximize cumulative rewards.
- Key Characteristics:
- - Involves agents, actions, states, and rewards.
- - Goal: Learn a policy for optimal decision-making.
- Applications:
- - Game Playing (e.g., Chess, Go)
- - Autonomous Vehicles
- - Robotics

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Types of Reinforcement Learning

- 1. Positive Reinforcement:
- - Reward for correct actions.
- - Example: Training a robot to walk.
- 2. Negative Reinforcement:
- - Penalty for incorrect actions.
- - Example: Learning to avoid obstacles in a maze.

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Comparision

- | Feature | Supervised Learning | Unsupervised Learning | Reinforcement Learning |
- |---|---|---|---|
- | Data Type | Labeled | Unlabeled | Interactive |
- | Goal | Predict output | Find patterns | Maximize rewards |
- | Applications | Image recognition, Spam segmentation | Customer | Game playing, Robotics |
- | Algorithms Examples | SVM, Decision Trees | K-Means, PCA | Q-Learning, Deep Q-Networks |

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